

UNIVERSITY OF THE ARMED FORCES ESPE

Object Oriented Programming

NRC: 3081

Jorge Edison Lascano

Members: Anthony Aimacaña - Kevin
Vinicio - Mateo Artieda

Enterprise Features Roster

Coffee Shop Management System

This document outlines the systematic breakdown of capabilities (Features) engineered into the platform. This matrix is structurally aligned with the Software Requirements Specification (SRS) and forms the foundation for all UML Use Case modeling.

Module 1: Point of Sale (POS) & Order Engine

Feature ID	Nomenclature	Technical Description
F1.1	Rapid Order Creation	Highly responsive touch interface engineered for sub-second product addition.
F1.2	Product Modifiers	Dynamic payload attachment for custom instructions (e.g., "No Sugar", "Oat Milk").
F1.3	CRM Integration	Modular association of transactions to registered Customer profiles or anonymous tokens.
F1.4	Real-Time Financials	Synchronous computation of Subtotals, localized Tax protocols, and Final Totals.
F1.5	Discount Authorization	Manager-authorized logic for applying percentage or fixed-amount price reductions.
F1.6	Offline Resilience	Local-cache fallbacks allowing uninterrupted order intake during network partitions.

Module 2: Checkout & Payment Gateway

Feature ID	Nomenclature	Technical Description
F2.1	Multi-Modal Parsing	Algorithmic routing for Cash, Credit/Debit, and direct Bank Transfer protocols.

F2.2	Tender Calculation	Automated precision calculation for cash change issuance.
F2.3	Tokenized Security	PCI-DSS compliant interactions via opaque Payment Tokens; zero raw card retention.
F2.4	Receipt Orchestration	Generation of standardized physical prints and digital (email) transactional receipts.

Module 3: Kitchen Display System (KDS)

Feature ID	Nomenclature	Technical Description
F3.1	State Transitions	Interactive toggles shifting entities through PENDING, PREPARING, and READY states.
F3.2	SLA Monitoring	Visual degradation/alerting for tickets exceeding Service Level Agreement time thresholds.

Module 4: Perpetual Inventory Control

Feature ID	Nomenclature	Technical Description
F4.1	Recipe Deduction	Algorithmic, micro-measurement subtraction of raw Ingredients upon order preparation.
F4.2	Threshold Alerts	Automated event generation when stock falls beneath pre-defined safety minimums.
F4.3	Preemptive Block	Real-time POS interlock preventing the sale of items lacking required raw materials.
F4.4	Audit & Reconciliation	Administrative GUI for manual stock overrides (handling shrinkage, spoilage, or refits).

Module 5: Floor & Human Capital Management

Feature ID	Nomenclature	Technical Description
F5.1	Spatial Mapping	State-machine tracking of physical table statuses (FREE, OCCUPIED).
F5.2	Staff Allocation	One-to-many relationship mapping between active Waiters and their assigned Tables.
F5.3	RBAC Enforcement	Strict Role-Based Access Control gating sensitive administrative actions.

Module 6: Administrative Backoffice

Feature ID	Nomenclature	Technical Description
F6.1		Comprehensive CRUD (Create, Read, Update, Delete) interfaces for system catalogs.
F6.2	Financial Reconciliation	End-of-Day (Z-Report) synthesis, aggregating revenue streams by payment type and operator.
F6.3	Velocity Analytics	High-level data aggregation detailing top-performing and under-performing SKUs.